

INDIAN KNOWLEDGE SYSTEMS (IKS)

HS271TA | Unit 2 – Structured Study Notes

Foundational Concepts of Mathematics and Science & Technology: Linguistics; Pāṇini's work on Sanskrit Grammar, Phonetics, Aṣṭādhyāyī's Vyākaraṇa • Number System and Units of Measurement: concept of zero, Piṅgala and the Binary system • Knowledge Framework & Classifications: Prameya, Pramāṇa, Saṃśaya

► UNIT AT A GLANCE

Part A – Linguistics: Pāṇini & Aṣṭādhyāyī, word generation, CFG mapping, Phonetics (Sthāna & Prayatna)

Part B – Number System: Zero, decimal system, mathematicians, large numbers, Bhūtasankhyā, Kaṭapayādi, Āryabhaṭa systems, Piṅgala & binary

Part C – Knowledge Framework: Pramāṇa Śāstra, Prameya, Pramāṇas (Pratyakṣa, Anumāna, Śabda, Upamāna), Saṃśaya

PART A – Linguistics: Pāṇini's Work on Sanskrit Grammar

1. Introduction to Pāṇini and the Aṣṭādhyāyī

In learning Sanskrit, the study of grammar plays a very important part. Among various Sanskrit grammatical systems, **Pāṇini's is the most significant and the most authoritative.**

- Pāṇini is best known for his seminal work, the **Aṣṭādhyāyī**, composed around the **5th or 6th century BCE**.
- The original work of Pāṇini's grammar is called **Aṣṭādhyāyī**. Previously, students of Sanskrit used to **commit to memory the whole work**.
- Pāṇini composed **3983 sūtras (rules)** to accommodate all the patterns and variations in the Sanskrit language.

Definition – Aṣṭādhyāyī: 'Ashta' = eight, 'adhyāya' = chapter. The work consists of **eight chapters**.

About Pāṇini

- Ancient Sanskrit **philologist, grammarian**, and revered scholar; born in the ancient province of **Gandhara**.
- His theory of **morphological analysis** was more advanced than any equivalent Western theory before the 20th century.
- Considered the **father of linguistics**.
- His grammar's focus on **brevity** gives it a highly unintuitive structure, reminiscent of modern notations such as the **Backus–Naur form (BNF)**.
- His contributions stemmed from a need to **preserve the correct interpretation of religious texts**. The grammar emphasizes **oral tradition** – students memorized the sūtras, often without written texts.
- The Aṣṭādhyāyī includes a unique **metalanguage**, enabling a precise description of Sanskrit that reflects both its **sounds and grammatical structures**.
- Birch-bark manuscript example: *Rūpāvatāra* – a grammatical textbook based on Pāṇini's grammar, composed by **Dharmakīrti**, a Buddhist monk from Ceylon (manuscript transcribed in 1663, Kashmir).

2. Lineage of Saṃskṛta Grammar – the Tri-Muni

- Aṣṭādhyāyī built upon **centuries of prior grammatical study**; provided a **standardized and universally accepted model**; surpassed earlier works due to **logical organization and precision**; marked the **high point** in India's grammatical tradition.

Sage	Period	Work / Role
Pāṇini	5th/6th c. BCE	Creator of Aṣṭādhyāyī – codification of grammar rules
Kātyāyana	4th c. BCE	Wrote Vārtikas – clarifying Pāṇini's sūtras
Patañjali	2nd c. BCE	Authored Mahābhāṣya – a monumental commentary

Together they are called the '**Tri-Muni**' (three sages) of Saṃskṛta grammar. [Unit 1 notes give Pāṇini c. 400 BCE – dates vary across sources.]

Structure of Aṣṭādhyāyī

- The **3983 rules** are arranged into **8 chapters**.
- Each chapter is further divided into **4 quarters (pādas)**.
- 8 chapters × 4 quarters = 32 sections**.

3. Rule-Based Nature of Saṃskṛta & Word Generation

- Language is an **assemblage of words**, combined in various ways to communicate ideas and transact knowledge. Understanding **word formation and generation** is central to language mastery.

Word formation is entirely rule-governed. The process involves:

- Starting with a **verbal or nominal root**
 - Adding appropriate **suffixes** to form words
- The correct word always emerges from correct rule application – this **eliminates ambiguity and ensures precision**.
 - Word derivation is **step-by-step and recursive**: at each step a rule is applied if conditions are satisfied; the process continues until **no more rules apply**, giving the final grammatically valid word.
 - The system is **modular**: base + suffix combinations create patterns across the vocabulary.

The Word Generation Scheme

Root type	Generation chain
Verbal root	Dhātu → (pratyaya) → kriyā-vibhaktayaḥ → kriyā-padam (verb word)
Nominal root	Prātipadika → (pratyaya) → nāma-vibhaktayaḥ → nāma-padam (noun word)
Conversion	A suffix can convert a verb to a noun (e.g., <i>Do</i> → <i>Doer</i>); converting noun to verb is also possible

Examples: karoti, kartā, putriyati, rāmaḥ, ramate

4. Example: Verbal Root 'kr' (to do)

[Source text writes the root as 'kru'; the Sanskrit root is *kr* (कृ).]

- In Pāṇini's **Dhātupāṭha** (list of verbal roots), the root is represented as '**Dukṛñ**' – Pāṇini attaches **indicators (it-markers) to the front and back** of verbal roots for mathematical transformation purposes.
- Removing 'du' and 'ñ' gives the substantive part of the verb, '**kṛ**' – something to do with *doing*. Example: **Karma** comes from kṛ.

Form	kṛ (to do)	paṭh (to read)	gam / gacch (to go)
Present (does)	karoti	paṭhati	gacchati
Continuous (doing)	kurvan	paṭhan	gacchan
Doer	kartā	paṭhitā	gantā
Having done	kṛtvā	paṭhitvā	gatvā
Please do (imperative)	karotu	paṭhatu	gacchatu
Must be done	kartavyam	paṭhitavyam	gantavyam
To do (infinitive)	kartum	paṭhitum	gantum
Done (past)	kṛtam	paṭhitam	gatam

★ KEY POINT

If you know the process for one verb, you can apply it to approximately **2200 verbs**.

Word Pattern Formation Logic

- '**kṛ**', '**paṭh**', '**gacch**' are called the **Base**. In Sanskrit you start with the Base, add a **suffix**, and get a word – but in the process some more rules are triggered.
- Example: verb '**kṛ**' + suffix '**tu**' → '**karotu**'.
- Word generation: start with a verbal root → add suffixes → add more suffixes → verb form → complete word.
- Noun root + suffix → noun form**. Some suffixes give masculine, some feminine.

The Numbers: Cases, Persons, Numbers

Category	Calculation	Result
Nominal root (e.g., <i>Rāma</i>)	7 cases × 3 numbers (singular, dual, plural)	21 suffixes → 21 word forms (e.g., adding 'su' to <i>Rāma</i> gives <i>Rāmaḥ</i>)
Verbal root	3 persons (1st, 2nd, 3rd) × 3 numbers = 9	9 additional suffix combinations (per tense/mood)

- Word generation is **algorithmic**: take the root, add suffixes on top of it, apply some rules, and derive the final word.

Dynamic Vocabulary and Modern Significance

- Vocabulary is **not fixed** – new words can be generated; as long as rules are followed, new valid forms are possible.
- Ensures **adaptability** of Saṃskṛta for changing needs; made it eternal and relevant across centuries; important for **lexicography and word creation**.

- Aṣṭādhyāyī resembles a **formal generative grammar** – rule-driven, logical, and **algorithmic**; highly suitable for **computer-based language processing**.
- Inspired modern fields such as **mathematics, computational linguistics, and AI**.
- Stands as the **world's oldest and most complete descriptive model of language**.

5. Computer Science View: Pāṇini and CFG

Simple Sentence Formation

Sanskrit: रामः फलं खादति (*Rāmaḥ phalam khādati* – 'Rama eats the fruit')

Word	Pāṇinian component
Rāmaḥ	noun – prātipadika + suP suffix → nominative singular
Phalam	noun – prātipadika + suP → accusative singular
Khādati	verb – dhātu + tiñ suffix → present tense, 3rd person singular

Equivalent CFG (context-free grammar):

```

S      → NP VP
NP     → N SUFFIX
VP     → V NP
N      → 'rāma' | 'phala'
SUFFIX → 'ḥ' | 'm'
V      → 'khādati'
```

Generated sentence: rāmaḥ phalam khādati. **Observation:** the CFG derivation mimics Pāṇini's derivational process of combining **stem + suffix** and **subject + object + verb**.

Compound Word Formation (Samāsa)

- Pāṇini: '**tatpuruṣaḥ samāsaḥ**' → a compound where one word depends on another. E.g., **Rāja-putraḥ** (king's son).
- Pāṇini-like rule: **Samāsa → N1 N2** [where N1 modifies N2]. CFG analogy: COMPOUND → N N; N → 'rāja' | 'putra'; Output: **rāja-putra**.

Conceptual Mapping: Pāṇini's Grammar vs CFG

Aspect	Pāṇini's Grammar	CFG	Mapping / Analogy
Basic Unit	<i>Sūtra</i> (rule)	Production rule	Each Pāṇinian sūtra acts like a CFG production
Non-terminals	Abstract categories (Prātipadika, Dhātu, Sup, Tiñ)	Non-terminal symbols (NP, VP)	Represent parts of speech or word forms
Terminals	Sounds or morphemes (phonemes, roots)	Terminal symbols	Smallest linguistic units (letters or tokens)
Derivation process	Sequential + recursive rule application (<i>Anuvṛtti</i>)	Recursive expansion of non-terminals	Both generate valid strings from abstract rules

Aspect	Pāṇini's Grammar	CFG	Mapping / Analogy
Rule type	Generative and transformative	Generative	CFG captures only the generative aspect (Pāṇini includes transformations too)
Recursion	Inherent in compound formation and syntax	Built-in	e.g., embedding of subordinate clauses
Meta-rules	<i>Paribhāṣā-sūtras</i> define how to interpret other rules	Meta-grammar (not in simple CFG)	Closer to meta-level CFG or attribute grammar

Learn further: Saṃsādhani – A Sanskrit Computational Toolkit (<https://sanskrit.uohyd.ac.in/scl/>) – includes a Sandhi-Splitter.

✓ SECTION SUMMARY

Aṣṭādhyāyī = **3983 sūtras, 8 chapters × 4 quarters**; Tri-Muni = Pāṇini, Kātyāyana, Patañjali.

Word = **root + suffix(es) + rules** (algorithmic, recursive); one pattern works for ~2200 verbs.

Nominal root → 7×3 = 21 forms; verbal root → 3×3 = 9 person-number suffixes.

Pāṇini's sūtras ≈ CFG production rules; Paribhāṣā-sūtras ≈ meta-rules.

Phonetics

6. Importance of Communication

"*idam andhaṃ tamaḥ kṛtsnaṃ jāyeta bhuvanatrāyāṃ | yadi śabdāhvayaṃ jyotiḥ ā saṃsārān na dīpyate ||*" – **Mahāsubhāṣitasamgraha**. Meaning: if the light called **śabda** did not shine, all three worlds would remain in darkness.

- Communication is key to **Trade, Science & Technology, Societal Progress**.
- Language is central to all human interactions and pursuits; effective processing of language ensures successful communication.

Dimensions of Language Processing

Mode	Activity	Meaning
Receptive (Input)	Listening & Reading	Understanding intended meaning
Productive (Output)	Speaking & Writing	Conveying ideas back to others

Language Processing Framework (2×2)

Perspective	Covers	Study area
Sound	Listening & Speaking	Phonetics
Script	Reading & Writing	Syntax & Grammar

Linguistics addresses all aspects of sound & script.

Definition – Linguistics: the **scientific study of language** (provides systematic methods for word forms & meaning) – covering **speech sounds, grammatical structures, meaning & syntax**.

Language and Vedic Tradition

- Preservation of the Vedas was considered a **sacred duty**. Vedāṅgas supported Vedic study; **three focus on language**:

Vedāṅga	Focus
Śikṣā	Phonetics (pronunciation & articulation)
Nirukta	Etymology (origin & meaning of words)
Vyākaraṇa	Grammar (rules of usage and structure)

- Mastery of Saṃskṛta ensured clear understanding of **epics, Purāṇas, and Vedas**.

7. Sounds: Sthāna (Place of Articulation) and Prayatna (Effort)

Place (Sthāna)	Name	Sounds
Throat	Kaṇṭha	अ क ख ग घ ङ ः ह
Palate	Tālu	इ च छ ज झ ञ य श
Upper palate	Mūrdhā	ऋ ॠ ऌ ॡ ढ ण र ष
Teeth	Dantāḥ	ल त थ द ध न ल स्
Lips	Oṣṭhau	उ प फ ब भ म्
Nasal	Nāsikā	ङ ञ ण न्म्

Place of pronunciation: Throat, Palate, Upper Palate, Teeth, Lips and Nasal. **Articulatory effort (Prayatna)** – *partial categorisation only*:

Category	Types of effort
Vowels	Length: Short, Long, Elongated Nasal / Non-nasal Pitch variations: Udātta, Anudātta, Svarita
Consonants	Alpa-prāṇa – mild breath Mahā-prāṇa – strong breath

✓ SECTION SUMMARY

Language processing = Sound (phonetics: listen/speak) + Script (grammar: read/write).

Three language Vedāṅgas: Śikṣā, Nirukta, Vyākaraṇa.

Sthāna = 6 places; Prayatna: vowels (length, nasality, pitch) and consonants (alpa/mahā-prāṇa).

PART B – Number System and Units of Measurement

"yathā śikhā mayūrāṇāṃ nāgānāṃ maṇayo yathā / tadvad vedāṅgaśāstrāṇāṃ gaṇitaṃ mūrdhni samsthitam
||"

★ KEY POINT

Just as the **crest on the heads of peacocks** and the **gems on the heads of serpents** are in the highest position, in the same way the **place of mathematics in the Vedāṅga-śāstras is at the top**.

1. Brief History of Indian Mathematics

Category	Names
Pioneering sages	Sūrya, Pitāmaha, Vaśiṣṭha, Medhātithi, Garga, Lomasa, Cyavana
Astronomer-mathematicians	Varāhamihira, Āryabhaṭa, Bhāskarācārya I & II, Brahmagupta, Mahāvīracārya, Kālakācārya, Pṛthuyasas, Pāvilūri Mallanna

- Invention of the **number system and concept of zero**; development of **basic operations** (addition, subtraction, multiplication, division); insights into the solar system and planetary motion; advances in algebra, calculus, geometry, astronomy (Post-Vedic period).

Global influence:

- Translations of Sanskrit texts into Arabic by **Kaṅka**; inspiration for **Al-Khwārizmī** and adoption of Sanskrit numerals.
- Spread to Spain, Germany, France, England, and beyond.
- Indian mathematics has a rich history going back **over 5,000 years**; influence spread to China, Southeast Asia, the Middle East, and Europe.
- Seminal contributions: **zero, geometry, arithmetic, binary mathematics, negative numbers, algebra, trigonometry, calculus**. The **decimal place value system** used worldwide today was first developed in India.
- **Classical period (400–1600 CE) = golden age** of Indian Mathematics: Aryabhata, Varahamihira, Brahmagupta, Bhaskara I, Mahavira, Bhaskara II, Madhava of Sangamagrama, Nilakantha Somayaji.

2. Concept of Zero (Śūnya)

'Zero' was discovered by an anonymous Indian. The numeral zero, as we understand it today, is an integral part of the **decimal numeral system**, developed in ancient India.

#	Aspect	Details
1	Early development	Earliest recorded use of a decimal system and zero traced to the Indus Valley Civilization (c. 2500–1500 BCE) ; zero as a number with its own value evolved later.
2	Brahmagupta (598–668 CE)	Defined zero as a number and discussed its properties; introduced rules for arithmetic operations with zero (addition, subtraction of zero) – in <i>Brāhmasphuṭasiddhānta</i> .
3	Āryabhaṭa (476–550 CE)	Used a symbol to represent zero and recognized its importance in calculations.

#	Aspect	Details
4	Spread	Zero and the decimal place-value system spread to the Islamic world (Al-Khwārizmī developed these ideas) and then to Europe in the Middle Ages.
5	Modern importance	A placeholder in the decimal system – represents numbers of varying magnitudes efficiently; crucial in algebra, calculus and other branches.

3. Number System in India

#	System / Period	Details
1	Indus Valley (c. 2500–1500 BCE)	Earliest known evidence of a numerical system; symbols for counting and recording quantities.
2	Brahmi numerals (c. 3rd century BCE)	Symbols for numbers – among the earliest written representations of numerical values in India.
3	Kharosthi numerals (3rd c. BCE – 3rd c. CE)	Positional decimal system (position determines value); symbols for 1–9 and a symbol for zero (as per notes); found on inscriptions, coins, artifacts.
4	Decimal system	India's most significant contribution; base ten, ten symbols (0–9) ; zero as placeholder and as a number; place-value notation simplifies arithmetic.
5	Invention of zero	Zero as both a placeholder and a number is attributed to Indian mathematicians; made complex calculations manageable.
6	Influence	Travelled to the Islamic world (Al-Khwārizmī) and eventually Europe → basis of modern mathematics.
7	Other numeral systems	Regional systems, e.g., Gupta numeral system (c. 4th–6th century CE); gradually replaced by the more efficient decimal system.

Origin of Number Systems in India

- Roots lie in the **Vedic period (c. 1500–500 BCE)**; the **Rigveda** has references to numbers and arithmetic operations used in rituals and measurements.
- Numerical concepts were embedded in religious practices, e.g., **construction of altars** with precise geometrical shapes and dimensions.
- Later Vedic **Śulba Sūtras** demonstrate advanced numerical skills – **fractions, proportions, square roots**.

Decimal System and Place Value Notation

- Earliest evidence of the decimal system: inscriptions from the **3rd century BCE (Ashokan Edicts)**.
- **Zero (śūnya)**, first formalized in India; **Brahmagupta** was the first to describe zero as a number and gave rules for it in *Brahmasphuṭasiddhānta*.
- Zero + place value = **infinite scalability** in numerical representation. Transmitted to the Islamic world via **Al-Khwārizmī and Al-Samaw'al**, then Europe.

4. Ancient Indian Mathematicians and Their Salient Contributions

#	Work / Mathematician	Period, Location	Salient Contributions
1	Vedic Texts	3000 BCE or earlier	Earliest recorded mathematical knowledge, number system, Pythagorean-type triplets ; decimal system of naming numbers; concept of infinity .
2	Śulba-sūtras – Baudhāyana, Āpastamba, Kātyāyana, Mānava	800–600 BCE	Earliest text of geometry ; approximate value of $\sqrt{2}$ and π ; exact procedures for construction/transformation of squares, rectangles, trapezia, etc.
3	Āryabhaṭa – <i>Āryabhaṭīyam</i>	476–550 CE; Kusumapura (near Pataliputra), Bihar	Concise verses; algorithms for square root, cube root ; place value system; sine table ; geometry; quadratic equations; linear indeterminate equations; sums of squares and cubes; planetary astronomy; plane and spherical trigonometry .
4	Varāhamihira – <i>Brhat Samhitā, Brhajjātaka, Pañcasiddhāntikā</i>	482–565 CE; Ujjain, Madhya Pradesh	Summary of five ancient siddhāntas ; sine table, trigonometric identities ($\sin^2\theta + \cos^2\theta = 1$); combinatorics ; magic squares .
5	Bhāskara I – Commentary on Āryabhaṭīya, <i>Laghubhāskarīyam, Mahābhāskarīyam</i>	600–800 CE; Vallabhi, Saurashtra, Gujarat	Expanded Āryabhaṭa's work on integer solutions of indeterminate equations ; approximate formula for the sine function ; planetary astronomy.
6	Brahmagupta – <i>Brāhmasphuṭasiddhānta, Khaṇḍakhadyaka</i>	598–668 CE; Bhillamala, Rajasthan	Rules of arithmetic with zero and negative numbers ; algebra (Bījagaṇita) ; linear and quadratic indeterminate equations; Pythagorean triplets; formula for diagonals and area of a cyclic quadrilateral ; notion of arithmetic mean.
7	Mahāvīracārya – <i>Gaṇitasārasaṅgraha</i>	800–870 CE; Gulbarga, Karnataka	First comprehensive, exclusive textbook on mathematics (arithmetic–geometry–algebra), continuing Jain mathematics tradition; permutations & combinations ; arithmetic and geometric series; sums of squares and cubes in AP.
8	Śrīpati – <i>Gaṇitatilaka, Siddhāntaśekhara</i> , etc.	1019–1066 CE; Rohinikhanda, Maharashtra	Planetary astronomy.
9	Bhāskarācārya (Bhāskara II) – <i>Līlāvātī</i> (arithmetic & geometry), <i>Bījagaṇita</i> (algebra), <i>Siddhānta-śiromaṇi</i> (astronomy), <i>Vāsanābhāṣya</i>	1114–1185 CE; Bijjadavida	Canonical textbooks used all over India; detailed explanations incl. upapatti (proof) ; addition formula for sine; surds ; permutations & combinations; indeterminate equations; ideas of calculus incl. mean value theorem ; planetary astronomy; construction of instruments.

#	Work / Mathematician	Period, Location	Salient Contributions
10	Kamalākara – <i>Siddhānta-tattvaviveka</i>	1616–1700 CE; Varanasi, UP	Addition and subtraction theorems for sine and cosine; sines and cosines of double, triple, etc. angles.

5. Large Numbers

The **Śukla Yajurveda (17.2)** has a list of names for powers of ten up to 10^{12} :

Name	Value	Name	Value
eka	1	prayuta	10^6
daśa	10	arbuda	10^7
śatam (mesochi)	10^2	nyarbuda	10^8
sahasra	10^3	saguran (samudra)	10^9
ayuta	10^4	madhya	10^{10}
niyuta	10^5	anta	10^{11}
parārdha	10^{12}		

[The source spells the 10^9 term "saguran"; it is commonly given as samudra in the Yajurveda list.]

- **Rigveda** period names: 1-ekam, 2-dve, 3-trīṇi, 4-catvāri, 5-pañca, 6-ṣaṭ, 7-sapta, 8-aṣṭa, 9-nava. Unique names for 10 to 100 in steps of ten: **daśa, viṃśati, triṃśat, catvāriṃśat, pañcāśat, ṣaṣṭi, sapṭati, aśīti, navati, śata**.
- Later Hindu and Buddhist texts extended this list, but lists are **not mutually consistent**; names for numbers larger than 10^8 differ between texts.
- E.g., **Pañcaviṃśa Brāhmaṇa**: 10^9 nikharva, 10^{10} vādava, 10^{11} akṣiti; **Śāṅkhāyana Śrauta Sūtra**: 10^9 nikharva, 10^{10} samudra, 10^{11} salila, 10^{12} antya, 10^{13} ananta.

Definition – Daśaguṇottara-saṃjñā: such lists of names for powers of ten.

- The entire universe measured in meters and then squared would be around **10^{53}** .
- In the **Lalitavistara-sūtra**, Siddhartha mentioned **tallakṣaṇa** as part of one counting system, and showed that **eight more counting systems** exist (each with 23 terms starting from the previous system); after nine successive systems he arrived at **10^{421}** (a 1 followed by 421 zeros!).

Text	Context	Number quoted
Lalitavistara-sūtra	Siddhartha during swayamvara	10^{421}
Kaccāyana's Pāli Grammar	—	10^{140}
Rāmāyaṇa	Size of Rama's army	10^{62}
Jain canonical works	Śīrṣaprahelikā – an estimate of time	$(8,400,000)^{28}$
Anuyogadvāra-sūtra	Jain canonical text	10^{28}
Taittirīya Upaniṣad	An inquiry into bliss	10^{21}

Text	Context	Number quoted
Līlāvātī	Description of place value number system	10^{17}

[Source table has merged cells in the last rows; values are listed here in the order given.]

6. Three Systems of Numeration

Definition – Numeration: the process of expressing numbers in **words, alphabets, sentences or verses**.

Well-defined number systems and units of measurement for length, weight, time and motion were used in Astronomy.

System	Principle
Bhūtasāṅkhyā	Represents numbers by equivalent (object) names ; large numbers represented using the decimal system
Kaṭapayādi	Uses ten symbols , each with a place and absolute value (letters → digits)
Āryabhaṭīya	Assigns a different absolute value to each alphabet

6.1 Bhūtasāṅkhyā System

Definition – Bhūtasāṅkhyā: '**bhūta**' (being) + '**sāṅkhyā**' (number) = '**number associated with a being**'.

Categories of entities used: Number names (Śūnya 0, Ekam 1, Dvi 2, Trīṇi 3, Catvāri 4...) • Physical entities (earth, moon, stars, mountain, fire, sky) • Animals (elephant = 8, horse, snake = 8) • Body parts (eyes, limbs, 7 Dhātus) • Gods (Śiva, Viṣṇu, Indra, Rāma = 3, Manu = 14, Agni = 3) • Concepts (seasons = 6, months = 12, days, 5 Bhūtas).

Number	Represented by (partial list)
0	Śūnya, Pūrṇa, Kha
1	Ādi, Candra, Pṛthivī, Eka
2	Aśvin, Pairs of limbs, Ayana, Dvandva, Dvi
3	Rāma, Guṇa, Loka, Kāla, Agni, Trinetra
4	Veda, Śruti, Yuga, Āśrama, Varṇa, Samudra, Kṛta
5	Bhūta, Śāstra, Bāṇa, Pāṇḍava, Indriya
6	Aṅga, Ṛtu, Darśana, Śaṇmukha, Ṣaṭ
7	Ṛṣi, Adri, Svara, Dhātu, Chandas
8	Vasu, Bhujaṅga, Siddhi, Dik, Kuñjara, Nāga
9	Gṛha, Aṅka, Nanda
10	Dik, Aṅgulī, Avatāra, Rāvaṇaśira
11	Rudra
12	Āditya, Rāśi

Number	Represented by (partial list)
13	Viśva, Kāma
14	Manu
15	Tithi, Dina

★ KEY POINT

Reading rule: "**Aṅkānāṃ vāmato gatiḥ**" – numbers are read from the **right side moving in reverse order** (the first word = units digit).

Examples (Bhūtasāṅkhyā)

#	Phrase	Digits (word order)	Number (reversed)
1	netra kha bhūmi śaśi sarpa	2, 0, 1, 1, 8	81102
2	veda-veda-aṅka-candraḥ	Veda 4, Veda 4, Aṅka 9, Candra 1	1944
3	kha-adri-rāma-agnayaḥ	Kha 0, Adri 7, Rāma 3, Agni 3	3370
4	bhujaṅga-nanda-dvi-naga-aṅga-bāṇa-ṣaṭ-kṛta-induvaḥ	8, 9, 2, 7 (naga = adri), 6, 5, 6, 4, 1	146,567,298

Mādhavācārya's approximation of π (Bhūtasāṅkhyā): "*nava-nikharva-mite vṛtta-vistare*" → nava = 9, nikharva = 10^{11} → diameter = 9×10^{11} . Circumference = 2,827,433,388,233.

$$\pi = \text{Circumference} / \text{Diameter} = 2827433388233 / (9 \times 10^{11}) \approx 3.14159265359$$

6.2 Kaṭapayādi System

The **ka-ṭa-pa-yā-di** system is an ancient Indian **alphasyllabic numeral system** that maps letters to numerals so that numbers can be remembered as **words or verses**.

- Assigns **more than one letter to one numeral** and **nullifies certain letters** as valueless – giving flexibility to form **meaningful words** out of numbers.
- A **mnemonic numeral system** encoding numbers into syllables (consonant + vowel units); used in Sanskrit (and later Malayalam, Kannada, Tulu) words and verses.

Rule verse (Sadratnamālā, Śaṅkaravarman): "nañāvacaśca śūnyāni saṅkhyāḥ kaṭapayādayaḥ | miśre tūpāntya-hal saṅkhyā na ca cintyo hala-svaraḥ ||"

[Standard reading: ña, na and the vowels (a-, etc.) = 0; in a conjunct, only the last consonant counts; a bare vowel carries no value. Numbers are read right-to-left (aṅkānāṃ vāmato gatiḥ).]

Digit	Letters (consonants)
1	क (ka), ट (ṭa), प (pa), य (ya)
2	ख (kha), ठ (ṭha), फ (pha), र (ra)
3	ग (ga), ड (ḍa), ब (ba), ल (la)
4	घ (gha), ढ (ḍha), भ (bha), व (va)

Digit	Letters (consonants)
5	ङ (ṅa), ण (ṇa), म (ma), श (śa)
6	च (ca), त (ta), ष (ṣa)
7	छ (cha), थ (tha), स (sa)
8	ज (ja), द (da), ह (ha)
9	झ (jha), ध (dha), ळ (ḷa)
0	ञ (ña), न (na)

Examples (Kaṭapayādi)

Word	Letters → digits	Reversed result
राघवाय (rāghavāya)	र = 2, घ = 4, व = 4, य = 1 → 2441	1442
भवति (bhavati)	भ = 4, व = 4, त = 6 → 446	644
Famous – π: भद्राम्बुद्धिसिद्धजन्मगणितश्रद्धा स्म यद् भूपगी	Relevant letters: भ(4) र(2) ब(3) ध(9) स(7) ध(9) ज(8) म(5) ग(3) ण(5) त(6) र(2) ध(9) म(5) य(1) भ(4) प(1) ग(3)	314159265358979324 → 3.14159265358979324 (π up to 17 decimals)
Melakartā ragas (Carnatic) – first two syllables encode the raga number	मेचकल्याणि – मे = 5, च = 6 → 65; वागदीश्वरी – वा = 4, ग = 3 → 34; गानमूर्ति – गा = 3, न = 0 → 3	Mechakalyāṇi = 65 , Vāgadīśvarī = 34 , Gānamūrti = 3

The first two syllables of the name of each **melakartā raga** are fitted so as to serve this formula – **72 Melakartas** in Carnatic music. Example in Śāstrīya raga: Swarajathi | Mechakalyani Raga | Adi Tala | *Sa Ni da pa pa ma pa ga ma*.

6.3 Āryabhaṭa System

"varga-akṣarāṇi varge 'varge 'varga-akṣarāṇi kāt ṇmau yaḥ | khadvina-ke svarā nava varge 'varge navāntya-vargevā ||" (Āryabhaṭīya)

It attributes a numerical value to each syllable of the form **consonant + vowel** possible in Sanskrit phonology, from **ka = 1** up to **hau = 10¹⁸**.

Group	Letters	Values
Varga consonants	क (ka) ... म (ma)	1 to 25 respectively (ka = 1, kha = 2 ... ma = 25)
Avarga consonants	य ra la va śa ṣa sa ha	ya = 30 , ra = 40 , la = 50 , va = 60 , śa = 70 , ṣa = 80 , sa = 90 , ha = 100

- Note: **य** takes the value of **ङम (ṅa-ma) = 5 + 25 = 30** (rule: ṇmau yaḥ).
- 9 vowels** are duplicated to make varga–avarga pairs; each vowel has a varga and avarga form and denotes the **place in the place-value system** (powers of ten):

Vowel	a / ā	i / ī	u / ū	ṛ	ḷ	e	o	ai	au
Power of 10	10^0	10^2	10^4	10^6	10^8	10^{10}	10^{12}	10^{14}	10^{16}

Rule	Meaning
वर्गाक्षराणि वर्गे	Varga letters (क to म) appear in varga places (even powers of 10: units, hundreds, etc.)
अवर्गेऽवर्गाक्षराणि	Avarga letters (य to ह) appear in avarga places (odd powers of 10: tens, thousands, etc.)
कात्	Letters are assigned values 1, 2, ... starting from क
झौ यः	ञ and म are equivalent to य (i.e., value 30)
खद्विनवके स्वराः नव वर्गेऽवर्गे नवान्त्यवर्गे वा	The nine vowels represent powers of ten and combine with consonants; they appear in both varga and avarga places and beyond if needed

Examples (Āryabhaṭa system)

Word	Calculation	Result
गुण (guṇa)	$gu = 3 \times 10^4 = 30000$; $ṇa = 15 \times 10^0 = 15$	30015
मखि (makhi)	$ma = 25 \times 10^0 = 25$; $khi = 2 \times 10^2 = 200$	225
ख्युघृ (khyughṛ)	$khyu = (2 + 30) \times 10^4 = 320000$; $ghṛ = 4 \times 10^6 = 4000000$	4,320,000 (= 1 Mahāyuga)

[Source writes $(2 \times 30) \times 10^4$ – it should be $(2 + 30) \times 10^4 = 320,000$ for the total 4,320,000 to hold.]

Yuga Cycle (4,320,000 solar years)

Yuga	Start (– End)	Length (solar years)	Divine years
Kṛta (Satya)	3,891,102 BCE	1,728,000	4,800
Tretā	2,163,102 BCE	1,296,000	3,600
Dvāpara	867,102 BCE	864,000	2,400
Kali	3102 BCE – 428,899 CE	432,000	1,200
Total		4,320,000	12,000

Basis	Bhūtasāṅkhyā	Kaṭapayādi	Āryabhaṭa
Principle	Objects/concepts stand for digits	Consonant letters → digits (many-to-one)	Each consonant has a value; vowels give place value (powers of 10)

Basis	Bhūtasāṅkhyā	Kaṭapayādi	Āryabhaṭa
Reading	Right to left (aṅkānāṃ vāmato gatiḥ)	Right to left	Sum of (consonant value × vowel power of 10)
Max range	Any (with large-number names)	Digit strings (e.g., π to 17 decimals)	ka = 1 ... hau = 10 ¹⁸
Uses	Astronomy, verse	Mnemonics, π, melakartā ragas	Astronomical numbers (e.g., yuga lengths)

✓ SECTION SUMMARY

Zero: Brahmagupta defined zero as a number + rules (598–668 CE); Āryabhaṭa used a zero symbol.
 Decimal place value (0–9) is India's greatest contribution; Ashokan edicts show it by 3rd c. BCE.
 Three numeration systems: **Bhūtasāṅkhyā**, **Kaṭapayādi**, **Āryabhaṭīya** – all read digits from the right (aṅkānāṃ vāmato gatiḥ) except the Āryabhaṭa sum method.
 Yuga cycle: 1,728,000 + 1,296,000 + 864,000 + 432,000 = **4,320,000**.

✦ MEMORY TRICK

Kaṭapayādi: **ka-ṭa-pa-ya = 1**, **kha-ṭha-pha-ra = 2**, **ga-ḍa-ba-la = 3** ... and so on; **ña & na = 0**. Read the digits right-to-left.

7. Piṅgala and the Binary System

Piṅgala, an Indian scholar of the **2nd century BCE**, contributed to the development of the **binary system**. While binary numbers are primarily associated with modern computing, his work on **Chandaḥ-śāstra** (a treatise on prosody) contains the **fundamental principles of the binary system**.

Chandaḥ-śāstra and the Hierarchical Structure of Meter

- Chandaḥ-śāstra deals with **prosody** – rules governing the meter and structure of poetic compositions.
- Meter has a **hierarchical structure**: **syllables → grouped into padas → combined to form the meter**.
- A **syllable** (basic building block of poetry) is a vowel, or a vowel with one or more consonants preceding it.

Laghu and Guru – the Binary Building Blocks

Type	Definition	Binary (as per notes)
Laghu (short)	Any syllable with a short vowel	1
Guru (long)	Any syllable with a long vowel A short syllable followed by a conjunct consonant A short syllable followed by an anusvāra or visarga The last syllable in a meter	0

Gaṇas – Binary Words of Length 3

- Groups of **three syllables** called **Gaṇas** – essentially **binary numbers of length 3**. There are **8 possible Gaṇas** (all combinations of Laghu and Guru, i.e., $2^3 = 8$). They were used to establish various rules for Chandaḥ-śāstra.
- Mnemonic: "yamātārājabhānasalagam"** (yamātā-rāja-bhāna-salagam) – each consecutive group of three syllables in the phrase represents a Gaṇa.

#	Gaṇa	Binary	Pattern (L = laghu 1, G = guru 0)
1	Ya gaṇa	100	L G G
2	Ma gaṇa	000	G G G
3	Ta gaṇa	001	G G L
4	Ra gaṇa	010	G L G
5	Ja gaṇa	101	L G L
6	Bha gaṇa	011	G L L
7	Na gaṇa	111	L L L
8	Sa gaṇa	110	L L G

Example – Bhagavad Gītā 4.7: *yadā yadā hi dharmasya glānir bhavati bhārata | abhyutthānam adharmasya tadātmānaṃ śṛjāmyaham* – the laghu/guru pattern and gaṇa division of the verse (33 akṣaras, 51 mātrās) is used to illustrate scansion.

Binary Cycle and de Bruijn Sequence

- The arrangement of Gaṇas in Piṅgala's mnemonic forms a **binary cycle of length 3**.
- This binary cycle is known as the **de Bruijn sequence** in **modern computer science**.

✓ SECTION SUMMARY

Piṅgala (2nd c. BCE) → Chandaḥ-śāstra → **Laghu (1) / Guru (0)** = binary digits.

8 Gaṇas (3-bit words) remembered via **yamātārājabhānasalagam**; arrangement = **de Bruijn sequence**.

PART C – Knowledge: Framework & Classifications

1. Prameya, Pramāṇa, Saṃśaya – Introduction

- All knowledge systems are founded on some **philosophical (metaphysical + epistemological + ethical) assumptions**. **IKS are no different**.
- "kāṇādaṃ pāṇinīyaṃ ca sarvaśāstropakāraṇam" – the Kāṇāda (Vaiśeṣika) and Pāṇinīya systems are useful to all śāstras. [Interpretation in bracket: *Vaiśeṣika provides the ontological/analytical framework, Pāṇinīya the linguistic framework.*]

2. Pramāṇa Śāstra (Epistemology)

Definition – Knowledge (as per Indian Philosophy): a **bank of true awareness acquired by a Mode of inquiry**.

The Knowledge Triangle

Element	Sanskrit	Meaning
Person	Pramātr	The knower / subject
Object	Prameya	The object of valid knowledge – what is to be known
Process	Pramāṇa	The means / process of acquiring valid knowledge
Centre	Pramā	Valid knowledge – arising from the interaction of the three

Nyāya Theory of Cognition – The Conception of Knowledge

Level	Branch	Sub-types
Jñāna (any awareness)	Anubhava (occurrent experience)	Prāmāṇya (True) → Pramā (knowledge) Aprāmāṇyam (False) → Doubt (saṃśaya), Error (viparyaya), Hypothetical reasoning (tarka)
	Smṛti (memory / disposition)	Prāmāṇya – True Aprāmāṇya – False

★ KEY POINT

Saṃśaya (doubt) is a type of *false/non-valid* cognition (along with viparyaya and tarka) – it marks **ambiguity in existing knowledge** that must be resolved to establish valid knowledge. [Prameya = what is known; Pramāṇa = how it is known.]

3. Pramāṇas – Means to Valid Cognition

Pramāṇas are the **instruments (strongest means) of acquiring true knowledge**.

Pramāṇa	Meaning
Pratyakṣa	Perception
Anumāna	Inference
Upamāna	Analogy
Śabda	Verbal Testimony

◆ MEMORY TRICK

P-A-U-Ś → Pratyakṣa, Anumāna, Upamāna, Śabda (the four Nyāya pramāṇas).

3.1 Pratyakṣa (Perception) Process

- Knowing an object by **direct contact** through the **5 external senses** – vision (cakṣu), hearing (śrotra), taste (rasana), smell (ghrāṇa), touch (tvak) – and the **6th sense, Manas (mind)**.

- Allows us to perceive **inner states** like happiness (sukha), sadness (duḥkha), desire (icchā), or even cognition itself (buddhi/jñāna).

"Ātmā manasā saṃyujyate | Manaḥ indriyeṇa, indriyam arthena | Tataḥ pratyakṣam |"

Ātman → Manaḥ → Indriya → Artha (Self → Mind → Sense organ → Object)

3.2 Anumāna (Inference)

Definition – Anumāna: the process by which we arrive at a cognition **based on another cognition** – knowledge that follows from prior understanding.

An inference has four logical elements:

Element	Meaning
1. Hetu	The sign / reason – the property which is known
2. Sādhya	The unknown – the property we want to infer
3. Vyāpti	The invariable concomitance relating the hetu and the sādhya
4. Pakṣa	Locus – the place where one wants to prove the sādhya

Pañca-Avayava Anumāna (Five-Step Reasoning Process)

Example: Force has been imparted to this ball, because it is moving.

Element	In the example
Sādhya	Having the property of being subjected to force
Pakṣa	The ball
Hetu	Having motion
Vyāpti	Wherever there is motion, there is the property of being subjected to force

Step	Name	Statement
1	Pratijñā (proposition)	The ball has the property of being subjected to force
2	Hetu (reason)	Because it has motion
3	Udāharaṇa (example)	Wherever there is motion, there is the property of being subjected to force, like a cart
4	Upanaya (application)	This ball is also so
5	Nigamana (conclusion)	Hence, the ball has the property of being subjected to force

✦ MEMORY TRICK

Five steps: **P-H-U-U-N** → **Pratijñā**, **Hetu**, **Udāharaṇa**, **Upanaya**, **Nigamana**.

3.3 Śabda (Verbal Testimony) – Knowledge through Language

āptopadeśaḥ śabdaḥ – **The assertion of a reliable person (āpta) is śabda** (Nyāya Sūtra 1.1.7).

- **Śabda** is the knowledge derived from the utterances (oral or written) of a **reliable source (āpta)**; it applies to both **Empirical Reality and Transcendental Realms**.
- **Āpta**: having **right knowledge, willingness to share, no intention to deceive, and free from attachment and aversion**.

3.4 Upamāna (Analogy)

Definition – Upamāna: the means of knowing the relationship between a **word and its meaning** through **similarity or comparison**.

Example – Gavaya:

1. A person unfamiliar with the word **Gavaya** (a wild animal) is told that it **resembles a cow**.
2. Later, upon seeing an animal (e.g., a **Nīlgai**) that resembles a cow, they **recognize it as a Gavaya**.
3. The key to upamāna is the **recognition of similarity**, which links the unknown term (gavaya) with the familiar concept (cow).

Pramāṇa	Basis	Example
Pratyakṣa	Direct sense–object contact (Ātman–Manas–Indriya–Artha)	Seeing a pot; feeling happiness
Anumāna	Reasoning via hetu–vyāpti–sādhya	Smoke → fire; moving ball → force
Upamāna	Similarity / comparison	Gavaya resembles a cow
Śabda	Word of a reliable person (āpta)	Scriptures, expert testimony

✓ SECTION SUMMARY

Knowledge triangle = **Pramātr (knower) + Prameya (object) + Pramāṇa (means) → Pramā (valid knowledge)**.

Four pramāṇas: **Pratyakṣa, Anumāna, Upamāna, Śabda**.

Anumāna needs **hetu, sādhya, vyāpti, pakṣa**; 5-step form = Pratijñā, Hetu, Udāharaṇa, Upanaya, Nigamana.

Samśaya (doubt), viparyaya (error), tarka (hypothetical reasoning) fall under aprāmāṇya (non-valid cognition).

Quick Revision Sheet

Topic	Must-remember facts
Aṣṭādhyāyī	3983 sūtras; 8 chapters × 4 quarters = 32; 5th/6th c. BCE; Pāṇini born in Gandhara
Tri-Muni	Pāṇini (Aṣṭādhyāyī), Kātyāyana (Vārtikas, 4th c. BCE), Patañjali (Mahābhāṣya, 2nd c. BCE)
Word generation	Root (dhātu/prātipadika) + suffix (pratyaya) + rules → pada; 7×3 = 21 nominal forms; 3×3 = 9 verbal suffixes; ~2200 verbs
Pāṇini vs CFG	Sūtra = production rule; Paribhāṣā-sūtra = meta-rule; Anuvṛtti = recursion/derivation

Topic	Must-remember facts
Phonetics	Sthāna: kaṇṭha, tālu, mūrdhā, danta, oṣṭha, nāsikā; Prayatna: alpa/mahā-prāṇa; udātta, anudātta, svarita
Zero	Brahmagupta (598–668 CE) – zero as a number; Āryabhaṭa (476–550 CE) – zero symbol
Large numbers	Yajurveda lists up to 10^{12} (parārdha); Lalitavistara 10^{421}
Bhūtasankhyā	Objects for digits; read right-to-left (aṅkānāṃ vāmato gatiḥ); netra-kha-bhūmi-śaśi-sarpa = 81102
Kaṭapayādi	ka-ṭa-pa-ya = 1 ...; ña, na = 0; rāghavāya = 1442; π to 17 decimals; 72 melakartā ragas
Āryabhaṭa system	ka–ma = 1–25; ya ... ha = 30–100; 9 vowels = 10^0 ... 10^{16} ; guṇa = 30015; hau = 10^{18}
Yuga cycle	Kṛta 1,728,000; Tretā 1,296,000; Dvāpara 864,000; Kali 432,000 = 4,320,000
Piṅgala	2nd c. BCE; Chandaḥ-śāstra; laghu 1 / guru 0; 8 gaṇas; yamātārājabhānasalagaṃ; de Bruijn sequence
Pramāṇa Śāstra	Pramāṭṛ – Prameya – Pramāṇa → Pramā
4 Pramāṇas	Pratyakṣa, Anumāna, Upamāna, Śabda
Anumāna	Hetu, Sādhya, Vyāpti, Pakṣa; 5 steps (Pratijñā ... Nigamana)
Āpta	Right knowledge, willingness to share, no intention to deceive, free from attachment & aversion